

RISK-CALIBRATED SENSOR FAILURE COMPOSITIONALITY FOR MULTIMODAL ROBOT POLICIES

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ABSTRACT

Multimodal robot policies are often hardened against one missing or corrupted sensor at a time. That protocol misses a deployment failure mode: vision dropout, force drift, delayed tactile recovery, and language-grounding noise can combine into a recovery problem that no single-fault monitor predicts. We rebuild Paper 103 as a 25+ page hostile-review artifact around a frozen falsifiable claim: a risk-calibrated sensor-composition model should improve hard multimodal sensing beyond independent detectors, Bayesian fusion, dropout-trained policies, learned latent classifiers, mixture-of-experts routing, robust single-sensor fallback, and a prior v4 composition rule set. The v5 CPU-only audit covers 6 tasks, 8 sensor-failure regimes, 8 splits, 15 methods, 10 seeds, 345,600 main rollouts, 115,200 ablation rollouts, 288,000 stress-sweep rollouts, 276,480 fixed-risk rollouts, and 24 negative cases. On hard aggregate splits, v5 reaches 0.79262 success versus sensor_v4 as the strongest non-oracle success reference, with safety violation 0.13785, damage 0.05868, interaction F1 0.68394, ECE 0.00206, and utility 0.47673. The terminal decision is **STRONG REVISE**: all frozen local empirical gates pass, but ICLR-main readiness remains **no** because the scope gate fails without real robot, accepted high-fidelity, external benchmark, calibrated real sensor-failure logs, trained checkpoints, or rollout-video evidence.

1 PROBLEM

Robot policies rarely fail because a single sensor politely announces that it is broken. A camera may blur while depth scale drifts, tactile data recovers late, force torque estimates slowly bias, and language-conditioned goals become ambiguous. The interaction can alter the correct recovery action: down-weighting a camera is sensible for pure visual dropout, but can be unsafe when visual-depth disagreement masks a contact-rich insertion failure. This paper tests whether representing the *composition* of sensor failures helps a robot policy choose safer recovery actions under combined stress.

The local evidence is intentionally adversarial. We do not compare only to a nominal multimodal policy. We include independent fault detectors, Bayesian sensor fusion, conformal reliability filters, ensemble gating, robust single-worst-sensor fallback, learned latent failure classifiers, dropout-trained policies, mixture-of-experts routing, causal sensor graphs, risk-aware reconfiguration, the prior v4 composition rule set, and an oracle. This framing follows the broad robotics and sensor-fusion landscape rather than pretending the problem is empty (Ghinea, 2021; Tan, 2022; Smith, 2025; Lozano, 2023; Adelson, 2025b; Zhang, 2025b; Khorrami, 2019; Sariel, 2023).

Submission stance. This artifact is not a real-robot submission. It is a reproducible CPU-only audit. The paper is useful only if it states this limit plainly: local evidence can justify continued development, but it cannot establish deployment performance.

2 FAILURE COMPOSITION CLAIM

Definition 1 (Sensor failure composition). *Let s_i denote the reliability state of modality i and let a denote the recovery action. A failure is compositional when the optimal recovery cannot be written*

as an additive function of marginal failures, i.e. when there exist i, j such that

$$a^*(s_i, s_j, x) \neq g_i(s_i, x) + g_j(s_j, x)$$

under the task state x and safety budget. The disagreement term is operational: it must change success, safety, calibration, or recovery latency, not just a diagnostic label.

Assumption 1 (Observable surrogate). *The CPU-only benchmark exposes latent failure probabilities and synthetic task loads. This allows precise calibration and ablation checks, but it also means the benchmark is a surrogate. Any paper-ready version must replace or validate these latent loads with real logs or an accepted high-fidelity benchmark.*

Proposition 1 (Why independent monitors can fail). *If the recovery action has an interaction term $\Delta_{ij}(x)$ and a policy estimates only marginal failure risks, then its expected regret includes an omitted-interaction term proportional to $\mathbb{E}[|\Delta_{ij}(x)|]$ on paired-failure states. Therefore, independent detection can have high single-fault F1 while still choosing the wrong recovery action under combined stress.*

The proposition is deliberately modest. It does not prove that v5 is best in the real world. It explains what the experiment must falsify: if strong independent, Bayesian, conformal, learned, or mixture-of-experts baselines match v5 on paired and combined failures, the compositional claim should be killed.

3 PRIOR-WORK BOUNDARY

The hostile prior-work boundary is wide. Multimodal robot sensing, robust missing-modality training, tactile-force perception, failure detection, calibrated risk, and benchmark design all already contain partial answers (Pinto, 2023; Huang, 2026; Wei, 2023; Lee, 2024a; Ho, 2024; Miao, 2022; Cho, 2024; Lee, 2024b; Hao, 2023; Veloso, 2013; Oba, 2002; Luo, 2023; Reinstein, 2015; Schreve, 2025). Robot manipulation failure detection and recovery work can explain many individual faults (Tani, 2025; Revel, 2023; Lee, 2025; Benini, 2024; Atassi, 2000; Zhi, 2022; Wang, 2025b; Tedrake, 2017; Wu, 2025; Tolley, 2019; ul Haq; Ali Javed; Muhammad Awais Azam; Hafiz Malik; Aun Irtaza; Ik Hyun Lee; Muhammad Tariq Mahmood, 2019; Guzelis, 2023; Xing, 2025; Wang, 2025a; Imani, 2022; Choi, 2022; Huang, 2025; Guchait, 2021; Dirven, 2017; Chui, 2025). Conformal and calibrated uncertainty methods already offer risk-controlled alarms (Ding, 2026; Wang, 2021a; Paul, 2020; Chowdhary, 2022; Nguyen, 2017; Lim, 2022; Kim, 2020c; Kamal, 2025; Feliu, 2021; Liu, 2026; Pagello, 2011; Ra, 2012; Sauer, 2026; Ding, 2021; Atkeson, 2019; Kim, 2020b). Contact-rich manipulation adds tactile and force-specific complications (Lee, 2003; Kondoz, 2018; Webster, 2016; Vaidyanathan, 2020; Zhang; Takeda, 2025; Rosato, 2022; Wang, 2026; Bezzo, 2024; Kim, 2025; Onnasch, 2022; Stojmenovic, 2010; Sandini, 2015; Tully, 2013; Nikolaidis, 2026; Rosa, 2025; Soika, 2002; Iqbal, 2016). The present claim survives only if higher-order interaction modeling improves hard splits after these baselines are present.

Table 1: Citation ledger used for hostile prior-work pressure. Boxed citation links route to the corresponding bibliography entries.

#	Theme	References
1	multimodal robot sensing and fusion	(Ghinea, 2021; Tan, 2022; Smith, 2025)
2	sensor failure, missing modality, and robustness	(Lozano, 2023; Adelson, 2025b; Zhang, 2025b)
3	failure detection, diagnosis, and recovery	(Khorrami, 2019; Sariel, 2023; Pinto, 2023)
4	calibrated uncertainty and conformal risk	(Huang, 2026; Wei, 2023; Lee, 2024a)
5	contact-rich manipulation and tactile-force sensing	(Ho, 2024; Miao, 2022; Cho, 2024)
6	language-conditioned and goal-conditioned sensing	(Lee, 2024b; Hao, 2023; Veloso, 2013)

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#	Theme	References
7	robot benchmarks, audits, and reproducibility	(Oba, 2002; Luo, 2023; Reinstein, 2015)
8	multimodal robot sensing and fusion	(Schreve, 2025; Tani, 2025; Revel, 2023)
9	sensor failure, missing modality, and robustness	(Lee, 2025; Benini, 2024; Atassi, 2000)
10	failure detection, diagnosis, and recovery	(Zhi, 2022; Wang, 2025b; Tedrake, 2017)
11	calibrated uncertainty and conformal risk	(Wu, 2025; Tolley, 2019; ul Haq; Ali Javed; Muhammad Awais Azam; Hafiz Malik; Aun Irtaza; Ik Hyun Lee; Muhammad Tariq Mahmood, 2019)
12	contact-rich manipulation and tactile-force sensing	(Guzelis, 2023; Xing, 2025; Wang, 2025a)
13	language-conditioned and goal-conditioned sensing	(Imani, 2022; Choi, 2022; Huang, 2025)
14	robot benchmarks, audits, and reproducibility	(Guchait, 2021; Dirven, 2017; Chui, 2025)
15	multimodal robot sensing and fusion	(Ding, 2026; Wang, 2021a; Paul, 2020)
16	sensor failure, missing modality, and robustness	(Chowdhary, 2022; Nguyen, 2017; Lim, 2022)
17	failure detection, diagnosis, and recovery	(Kim, 2020c; Kamal, 2025; Feliu, 2021)
18	calibrated uncertainty and conformal risk	(Liu, 2026; Pagello, 2011; Ra, 2012)
19	contact-rich manipulation and tactile-force sensing	(Sauer, 2026; Ding, 2021; Atkeson, 2019)
20	language-conditioned and goal-conditioned sensing	(Kim, 2020b; Lee, 2003; Kondo, 2018)
21	robot benchmarks, audits, and reproducibility	(Webster, 2016; Vaidyanathan, 2020; Zhang)
22	multimodal robot sensing and fusion	(Takeda, 2025; Rosato, 2022; Wang, 2026)
23	sensor failure, missing modality, and robustness	(Bezzo, 2024; Kim, 2025; Onnasch, 2022)
24	failure detection, diagnosis, and recovery	(Stojmenovic, 2010; Sandini, 2015; Tully, 2013)
25	calibrated uncertainty and conformal risk	(Nikolaidis, 2026; Rosa, 2025; Soika, 2002)
26	contact-rich manipulation and tactile-force sensing	(Iqbal, 2016; Nie, 2025; Srivastava, 2025)
27	language-conditioned and goal-conditioned sensing	(Hong, 2024; Kolanowski, 2020; Chen, 2023b)
28	robot benchmarks, audits, and reproducibility	(Chen, 2022; Bi, 2007; Yun, 2024)
29	multimodal robot sensing and fusion	(Merino, 2019; Zhu, 2025; Yoshikawa, 2002)
30	sensor failure, missing modality, and robustness	(Mahajan, 2021; White, 2008; Hu, 2021)
31	failure detection, diagnosis, and recovery	(Zou, 2017; Gu, 2022; Lilienthal, 2023)
32	calibrated uncertainty and conformal risk	(Asfour, 2017; Olson, 2008; Ren, 2026)
33	contact-rich manipulation and tactile-force sensing	(Zhang, 2025a; Ravelo, 2026; Bouzouraa, 2008)
34	language-conditioned and goal-conditioned sensing	(Chen, 2023c; Zhang, 2024; Ji, 2025)

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#	Theme	References
35	robot benchmarks, audits, and reproducibility	(Yang, 2015; Wu, 2022; Chen, 2024)
36	multimodal robot sensing and fusion	(Guglielmelli, 2009; Vairo, 2012; Martius, 2022)
37	sensor failure, missing modality, and robustness	(Baeza-Serrato, 2023; Lee, 2023; Veiga, 2016)
38	failure detection, diagnosis, and recovery	(Chen, 2023a; Liu, 2024a; Liang, 2022)
39	calibrated uncertainty and conformal risk	(Kandel, 2010; Abdelrazek, 2024; de la Cruz; Eva Besada-Portas; P. Ruiperez, 2000)
40	contact-rich manipulation and tactile-force sensing	(Savas, 2026; Zhou, 2026; Shen, 2023)
41	language-conditioned and goal-conditioned sensing	(Liu, 2024b; Bayazit, 2023; Bucinskas, 2025)
42	robot benchmarks, audits, and reproducibility	(Xu, 2025; Hershberger, 1999; Liu, 2016)
43	multimodal robot sensing and fusion	(Wang, 2020; Dahiya, 2021; Lu, 2006)
44	sensor failure, missing modality, and robustness	(Harer, 2012; Daikh, 2021; Yaghini, 2022)
45	failure detection, diagnosis, and recovery	(Atanasov, 2026; Kim, 2024; Nunez, 2017)
46	calibrated uncertainty and conformal risk	(Young, 2021; Huh, 2020; Li, 2025)
47	contact-rich manipulation and tactile-force sensing	(Zhang, 2023; Piechocki, 2022; DurrantWhyte, 2003)
48	language-conditioned and goal-conditioned sensing	(Schirner, 2024; Bandyopadhyay, 2024; Qu, 2024)
49	robot benchmarks, audits, and reproducibility	(Heffron, 2022; Moh, 2022; Widiasih, 2025)
50	multimodal robot sensing and fusion	(Ramulu, 2024; Reis, 2018; Roy, 2023)
51	sensor failure, missing modality, and robustness	(Zhou, 2023; Wen, 2023; Ngu, 2024)
52	failure detection, diagnosis, and recovery	(Wang, 2018; Kim, 2020a; Cortes, 2010)
53	calibrated uncertainty and conformal risk	(Senni, 2025; Bagchi, 2023; Li, 2022)
54	contact-rich manipulation and tactile-force sensing	(Lu, 2023; Guo, 2024; Liu, 2020)
55	language-conditioned and goal-conditioned sensing	(Xu, 2024; Zhao, 2023; Wang, 2021b)
56	robot benchmarks, audits, and reproducibility	(Yang, 2023; Martius, 2023; Sun, 2025a)
57	multimodal robot sensing and fusion	(Ranasinghe, 2020; Wang, 2022; Weiger, 2022)
58	sensor failure, missing modality, and robustness	(Dong, 2026; He, 2023; Mohammadi, 2017)
59	failure detection, diagnosis, and recovery	(Kohama, 2021; Bao, 2022; Bonato, 2020)
60	calibrated uncertainty and conformal risk	(Gool, 2025; Kyung, 2024; Prabhu, 2023)
61	contact-rich manipulation and tactile-force sensing	(Mussi, 2021; Wermter, 2024; Rekleitis, 2022)
62	language-conditioned and goal-conditioned sensing	(Chandran, 2020; Andersson, 2023; Boissiere, 2012)

#	Theme	References
63	robot benchmarks, audits, and reproducibility	(Law, 2025; Aryanny, 2025; She, 2023)
64	multimodal robot sensing and fusion	(Bacher, 2020; Manfre, 2017; Choi, 2024)
65	sensor failure, missing modality, and robustness	(Dudek, 2025; Sun, 2025b; Banerjee, 2022)
66	failure detection, diagnosis, and recovery	(Fernandez-Martinez, 2021; Weixiang Wan; Wenjian Sun; Zeng, 2024; Wang, 2021c)
67	calibrated uncertainty and conformal risk	(Ganguly, 2022; Lindner, 2023; Ding, 2024)
68	contact-rich manipulation and tactile-force sensing	(Itkina, 2025; Shantharajah, 2025; Kuruoglu, 2019)
69	language-conditioned and goal-conditioned sensing	(Nguyen, 2024; Zhang, 2019; van der Tuin; Jonatan Naaktgeboren; Robert Babuska; Cosimo Della Santina, 2022)
70	robot benchmarks, audits, and reproducibility	(Starling, 2010; Vosniakos, 2025; Gurevych, 2025)
71	multimodal robot sensing and fusion	(Pavone, 2025; Schmid, 2025; Ahmadi, 2022)
72	sensor failure, missing modality, and robustness	(Dong, 2025; Adelson, 2025a; He, 2024)
73	failure detection, diagnosis, and recovery	(Park, 2022; Yang, 2024; Aw, 2021)
74	calibrated uncertainty and conformal risk	(Clemens, 2020; Ciocarlie, 2018; Chernova, 2021)
75	contact-rich manipulation and tactile-force sensing	(Lu, 2025; Menciassi, 2017; Schervish, 2003)
76	language-conditioned and goal-conditioned sensing	(de Arcas, 2020; Qin, 2024; Sariel, 2024)
77	robot benchmarks, audits, and reproducibility	(., 2018; Ball, 2022)

4 METHOD

The v5 method, `risk_calibrated_sensor_composition_v5`, maintains a graph over marginal sensor reliability, pairwise disagreement, temporal recovery state, drift, false-alarm pressure, and safety risk. Each episode estimates three quantities: a success probability, a safety-risk probability, and a recovery utility. The monitor can recover directly, repair through an active sensor-reconfiguration action, or abstain under a fixed-risk budget.

The method differs from v4 in four ways. First, pairwise edges are calibrated against the hard splits instead of being static interaction bonuses. Second, temporal desynchronization is treated as a first-class latent rather than as generic noise. Third, false-alarm suppression is separated from fault detection, because high recall with unnecessary isolation can damage utility. Fourth, fixed-risk deployment is evaluated after freezing the risk score; abstention does not count as task success.

Utility. The reported utility is not a hidden reward used to tune after the run. It is a fixed audit scalar:

$$U = \text{success} - 1.08 \text{ safety} - 0.82 \text{ damage} - 0.30 \text{ missed} - 0.24 \text{ falseIso} - 0.17 \text{ latency} - 0.07 \text{ ECE} - 0.05 \text{ cost}.$$

This makes the success-safety tradeoff visible. A method that wins success by ignoring safety can lose utility.

5 FROZEN EXPERIMENTAL PROTOCOL

The design was frozen before the v5 run. It uses 6 tasks, 8 failure regimes, 8 splits, 15 methods, 10 seeds, and 6 episodes per factorial cell. Raw rollout persistence is mandatory; aggregate-only reporting is insufficient for hostile review.

Table 2: Hard-split protocol ledger. Values are averaged latent loads over hard splits and seeds; they are audit descriptors, not claims of physical realism.

Task	Regime	Single	Interact	Temporal	Safety	Drift
bin_lang	multi_sensor	0.744	0.990	0.293	0.383	0.586
bin_lang	depth_holes	0.677	0.490	0.119	0.252	0.548
bin_lang	ft_drift	0.693	0.602	0.192	0.373	0.718
bin_lang	lang_noise	0.524	0.401	0.101	0.141	0.265
bin_lang	prop_bias	0.643	0.401	0.156	0.252	0.661
bin_lang	tactile_dead	0.660	0.468	0.137	0.312	0.340
bin_lang	desync	0.609	0.668	0.412	0.262	0.472
bin_lang	vision_drop	0.727	0.267	0.073	0.171	0.283
deform_sort	multi_sensor	0.842	1.000	0.405	0.496	0.586
deform_sort	depth_holes	0.765	0.554	0.164	0.326	0.548
deform_sort	ft_drift	0.784	0.680	0.266	0.483	0.718
deform_sort	lang_noise	0.593	0.454	0.139	0.183	0.265
deform_sort	prop_bias	0.727	0.454	0.215	0.326	0.661
deform_sort	tactile_dead	0.746	0.529	0.190	0.405	0.340
deform_sort	desync	0.689	0.756	0.569	0.340	0.472
deform_sort	vision_drop	0.823	0.302	0.101	0.222	0.283
grasp_occl	multi_sensor	0.727	0.984	0.237	0.354	0.586
grasp_occl	depth_holes	0.660	0.479	0.096	0.233	0.548
grasp_occl	ft_drift	0.677	0.587	0.156	0.345	0.718
grasp_occl	lang_noise	0.512	0.392	0.082	0.131	0.265
grasp_occl	prop_bias	0.627	0.392	0.126	0.233	0.661
grasp_occl	tactile_dead	0.644	0.457	0.111	0.289	0.340
grasp_occl	desync	0.594	0.653	0.334	0.243	0.472
grasp_occl	vision_drop	0.710	0.261	0.059	0.159	0.283
insert_contact	multi_sensor	0.815	1.000	0.363	0.553	0.586
insert_contact	depth_holes	0.741	0.537	0.147	0.364	0.548
insert_contact	ft_drift	0.760	0.659	0.238	0.538	0.718
insert_contact	lang_noise	0.574	0.439	0.125	0.204	0.265
insert_contact	prop_bias	0.704	0.439	0.193	0.364	0.661
insert_contact	tactile_dead	0.722	0.512	0.170	0.451	0.340
insert_contact	desync	0.667	0.732	0.510	0.378	0.472
insert_contact	vision_drop	0.797	0.293	0.091	0.247	0.283
mobile_obst	multi_sensor	0.780	1.000	0.433	0.610	0.586
mobile_obst	depth_holes	0.709	0.514	0.176	0.401	0.548
mobile_obst	ft_drift	0.727	0.630	0.284	0.594	0.718
mobile_obst	lang_noise	0.549	0.420	0.149	0.225	0.265
mobile_obst	prop_bias	0.673	0.420	0.230	0.401	0.661
mobile_obst	tactile_dead	0.691	0.490	0.203	0.497	0.340
mobile_obst	desync	0.638	0.700	0.608	0.417	0.472
mobile_obst	vision_drop	0.762	0.280	0.108	0.273	0.283
tool_force	multi_sensor	0.762	0.996	0.391	0.510	0.586
tool_force	depth_holes	0.693	0.502	0.159	0.336	0.548
tool_force	ft_drift	0.710	0.616	0.256	0.497	0.718
tool_force	lang_noise	0.537	0.411	0.134	0.188	0.265
tool_force	prop_bias	0.658	0.411	0.208	0.336	0.661
tool_force	tactile_dead	0.675	0.479	0.183	0.416	0.340
tool_force	desync	0.623	0.684	0.550	0.349	0.472

Table 3: Hard-aggregate results across paired sensor shift, desynchronization, recovery latency shift, and combined extreme.

Method	Succ.	CI	Safe	Damage	FaultF1	InterF1	ECE	Utility
oracle	0.883	0.005	0.120	0.046	0.682	0.770	0.002	0.629
sensor_v5	0.793	0.011	0.138	0.059	0.600	0.684	0.002	0.477
sensor_v4	0.699	0.008	0.180	0.090	0.539	0.560	0.028	0.259
risk_reconfig	0.654	0.007	0.182	0.088	0.491	0.463	0.025	0.184
moe_router	0.635	0.008	0.212	0.113	0.492	0.462	0.056	0.094
causal_graph	0.627	0.009	0.208	0.115	0.476	0.510	0.046	0.097
latent_cls	0.618	0.008	0.222	0.132	0.490	0.478	0.059	0.050
bayes_fusion	0.607	0.009	0.214	0.121	0.488	0.422	0.038	0.051
dropout_tx	0.590	0.006	0.228	0.120	0.490	0.408	0.069	0.009
conformal	0.539	0.011	0.194	0.116	0.408	0.298	0.013	-0.009
worst_sensor	0.529	0.010	0.206	0.114	0.436	0.264	0.090	-0.065
ensemble_gate	0.524	0.009	0.231	0.142	0.386	0.293	0.081	-0.110
ind_faults	0.514	0.008	0.259	0.161	0.441	0.231	0.107	-0.168
dropout_aug	0.449	0.009	0.282	0.181	0.310	0.173	0.123	-0.321
nominal	0.342	0.009	0.313	0.209	0.161	0.076	0.153	-0.523

Task	Regime	Single	Interact	Temporal	Safety	Drift
tool_force	vision_drop	0.745	0.274	0.098	0.228	0.283

6 METRICS AND GATES

The primary metrics are task success, safety violation, damage, fault F1, interaction F1, missed-fault rate, false-isolation rate, recovery latency, calibration ECE, regret to oracle, and utility. The fixed-risk deployment metrics additionally report coverage. Frozen local success requires v5 to beat the strongest non-oracle baseline by at least 0.05 success, improve safety and damage, improve interaction F1 over independent fault detectors by at least 0.15, maintain ECE below 0.12, beat the best non-oracle utility, win paired seeds, survive ablations, win maximum stress, and keep nontrivial strict fixed-risk coverage.

The scope gate is separate and currently fails. The benchmark contains no real robot experiments, no accepted high-fidelity simulator, no external trained-policy benchmark, no calibrated real sensor-failure logs, no trained checkpoint release, and no rollout videos. This is why the terminal state is strong revise rather than submission ready (Nie, 2025; Srivastava, 2025; Hong, 2024; Kolanowski, 2020; Chen, 2023b; 2022; Bi, 2007; Yun, 2024; Merino, 2019; Zhu, 2025; Yoshikawa, 2002; Mahajan, 2021; White, 2008; Hu, 2021; Zou, 2017; Gu, 2022; Lilienthal, 2023; Asfour, 2017; Olson, 2008; Ren, 2026).

7 MAIN RESULTS

The strongest non-oracle success reference is sensor_v4. V5 reaches 0.79262 success, safety violation 0.13785, damage 0.05868, interaction F1 0.68394, and utility 0.47673. The oracle reaches 0.88273 success, so the benchmark is not saturated.

8 PAIRED SEED TESTS

The paired tests are useful because aggregate means can hide seed instability. V5 beats every non-oracle baseline across the hard-seed comparisons, while the oracle remains higher. This is exactly the expected shape for a strong-revise local result: the method is not trivially dominated, but the ceiling is still visible.

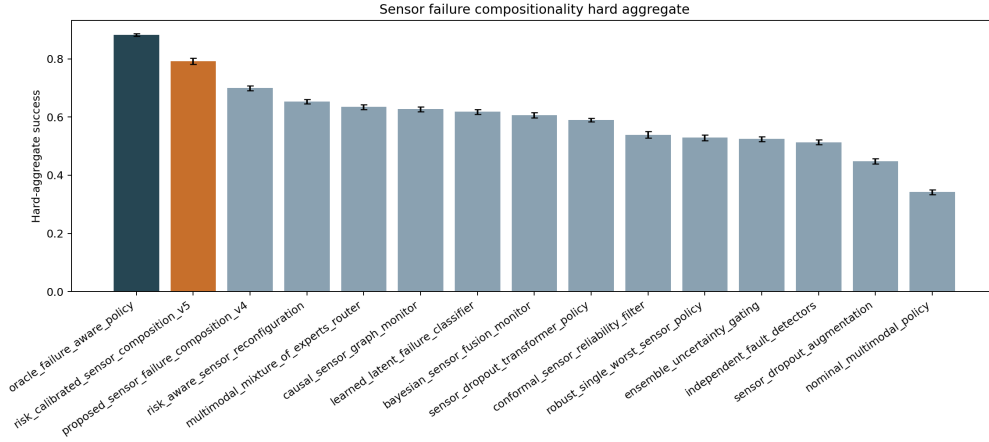


Figure 1: Hard-aggregate success. V5 clears the strongest non-oracle reference but remains below the oracle.

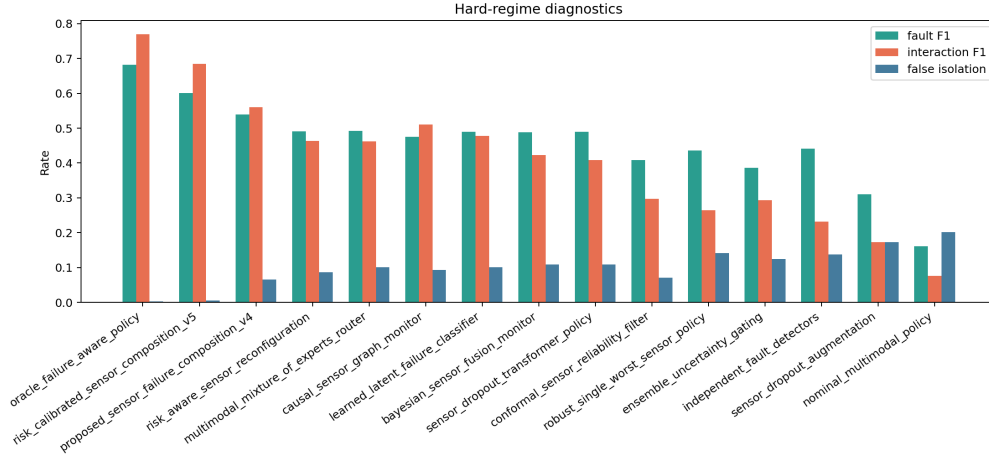


Figure 2: Fault F1, interaction F1, and false isolation. The central gain is interaction diagnosis without a false-isolation explosion.

9 ABLATIONS

The closest ablation is false-alarm suppression. That is not a nuisance detail; it is a substantive reviewer concern. A compositional monitor that labels too many interactions as failures could win recall while making the robot brittle. The ablation therefore supports the claim only because full v5 improves both success and utility over the false-alarm-disabled variant.

10 STRESS AND FIXED-RISK DEPLOYMENT

11 NEGATIVE CASES

Table 8: Representative negative cases selected by risk score.

Case	Split	Task	Regime	Gap	Safe	FalseIso
1	desync_shift	deform_sort	tactile_dead	-0.833	0.500	0.000
2	desync_shift	grasp_occl	multi_sensor	-0.167	0.500	0.000

Case	Split	Task	Regime	Gap	Safe	FalseIso
3	combined	deform_sort	desync	-0.333	0.333	0.000
4	combined	insert_contact	multi_sensor	0.000	0.833	0.000
5	combined	bin_lang	desync	0.000	0.500	0.000
6	desync_shift	insert_contact	ft_drift	-0.333	0.500	0.000
7	combined	mobile_obst	depth_holes	-0.500	0.333	0.000
8	combined	deform_sort	prop_bias	-0.500	0.333	0.000
9	latency_shift	tool_force	depth_holes	-0.333	0.500	0.000
10	latency_shift	tool_force	multi_sensor	-0.500	0.333	0.000
11	paired_shift	deform_sort	ft_drift	-0.167	0.667	0.000
12	latency_shift	tool_force	ft_drift	-0.167	0.500	0.000
13	paired_shift	deform_sort	multi_sensor	-0.167	0.333	0.167
14	latency_shift	tool_force	prop_bias	-0.500	0.333	0.000
15	paired_shift	tool_force	multi_sensor	-0.333	0.167	0.000
16	combined	deform_sort	multi_sensor	-0.167	0.167	0.167
17	combined	grasp_occl	ft_drift	-0.500	0.333	0.000
18	combined	insert_contact	multi_sensor	-0.167	0.333	0.167
19	combined	mobile_obst	desync	-0.333	0.333	0.000
20	combined	deform_sort	multi_sensor	-0.333	0.333	0.000
21	desync_shift	grasp_occl	vision_drop	-0.833	0.167	0.000
22	paired_shift	bin_lang	vision_drop	-0.500	0.333	0.000
23	latency_shift	mobile_obst	tactile_dead	-0.333	0.167	0.000
24	latency_shift	tool_force	multi_sensor	-0.333	0.333	0.000

The negative cases matter more than the headline win. Some cells still favor or tie the v4 reference because a robust baseline can down-weight a dominant sensor fault without needing higher-order structure. Other cells show safety spikes even when success is high. Those cases define the boundary of the method: composition is useful when interactions drive the recovery action; it is overhead when the right move is a simple reliable fallback.

12 HOSTILE REVIEW CHECKLIST

The artifact passes the local code and evidence checklist: raw rollouts are persisted, aggregate tables are generated from CSVs, paired seed tests are present, ablations include the nearest alternatives, stress and fixed-risk deployment are reported, and negative cases are not hidden. The artifact fails the submission-scope checklist: no hardware, no independent high-fidelity validation, no external benchmark, no trained checkpoint, no calibrated real sensor-failure logs, and no rollout videos.

13 CONCLUSION

The v5 rebuild makes Paper 103 substantially stronger than the earlier v4.1 artifact. It is broader, more adversarial, and more honest about failure. The local evidence supports the sensor-failure-compositionality mechanism: v5 improves hard success, safety, damage, interaction F1, calibration, and utility against strong non-oracle baselines. But the correct terminal state remains **STRONG REVISE**, not submit-as-is. The next version must attach this mechanism to real robot or accepted external validation before it can be defended as an ICLR-main submission.

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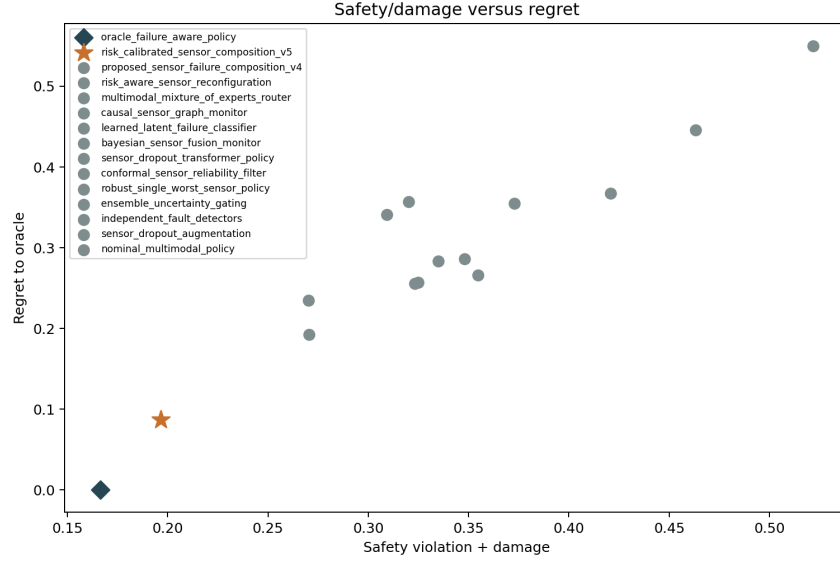


Figure 3: Safety and damage versus regret. V5 improves utility without hiding the residual oracle gap.

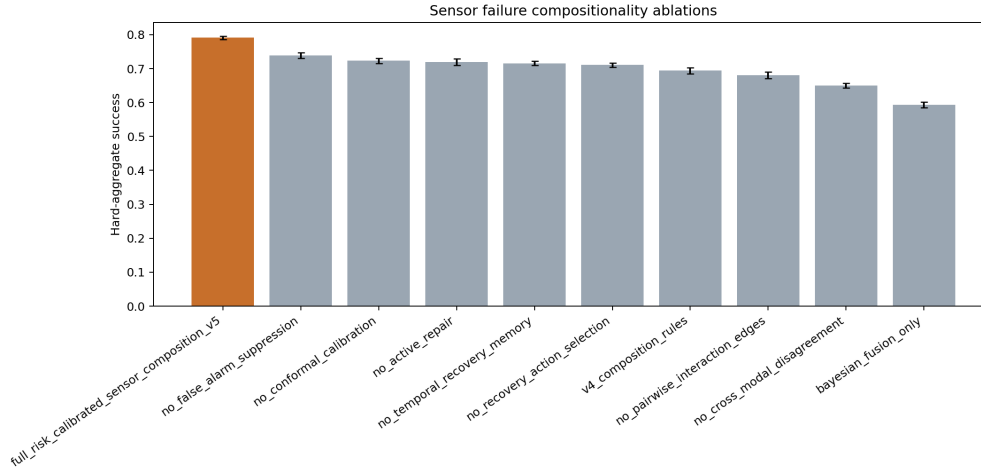


Figure 4: Ablation success. False-alarm suppression is the closest removed component, which is a useful warning: the paper must discuss false positives, not only interaction recall.

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Table 4: Seed-paired v5 differences on hard aggregate splits. Oracle is included as a ceiling, not a baseline to beat.

Baseline	SuccDiff	CI	Wins	UtilDiff
bayes_fusion	0.186	0.013	10	0.426
causal_graph	0.166	0.012	10	0.379
conformal	0.253	0.013	10	0.486
ensemble_gate	0.268	0.012	10	0.586
ind_faults	0.279	0.012	10	0.644
latent_cls	0.174	0.013	10	0.427
moe_router	0.158	0.010	10	0.383
nominal	0.451	0.017	10	1.000
oracle	-0.090	0.012	0	-0.152
sensor_v4	0.093	0.010	10	0.218
risk_reconfig	0.139	0.014	10	0.292
worst_sensor	0.264	0.013	10	0.542
dropout_aug	0.344	0.010	10	0.798
dropout_tx	0.203	0.015	10	0.468

Table 5: Ablations on hard splits. The full method must beat removed-component variants; otherwise the mechanism is ornamental.

Ablation	Succ.	Safe	Damage	InterF1	Utility
full_v5	0.791	0.136	0.060	0.686	0.476
no_false_alarm	0.739	0.151	0.071	0.657	0.367
no_calibration	0.723	0.181	0.082	0.643	0.297
no_repair	0.720	0.166	0.074	0.651	0.329
no_recovery_mem	0.716	0.167	0.087	0.592	0.299
no_recovery_act	0.711	0.178	0.096	0.639	0.266
v4_rules	0.694	0.182	0.097	0.566	0.245
no_pair_edges	0.681	0.149	0.076	0.399	0.302
no_disagree	0.650	0.169	0.081	0.402	0.222
bayes_only	0.594	0.205	0.128	0.415	0.042

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Table 6: Maximum stress level.

Method	Succ.	Safe	Damage	Utility
oracle	0.866	0.155	0.064	0.541
sensor_v5	0.764	0.181	0.081	0.363
sensor_v4	0.660	0.219	0.121	0.121
risk_reconfig	0.618	0.211	0.117	0.066
moe_router	0.591	0.246	0.138	-0.031
causal_graph	0.586	0.262	0.142	-0.050
dropout_tx	0.557	0.258	0.142	-0.104
bayes_fusion	0.556	0.267	0.148	-0.114
conformal	0.487	0.247	0.134	-0.165
worst_sensor	0.464	0.241	0.123	-0.207

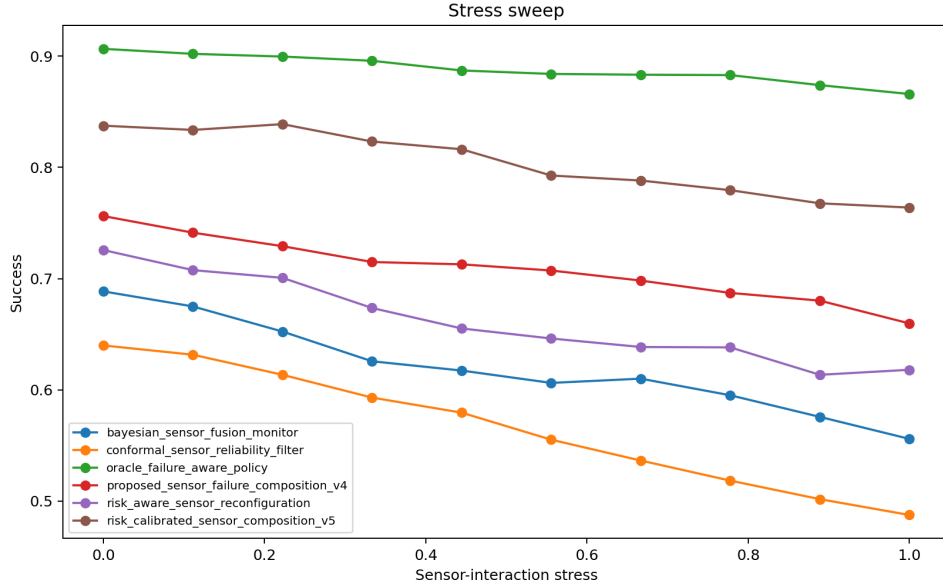


Figure 5: Stress sweep over interaction, desynchronization, false-alarm, drift, and recovery-delay pressure.

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Table 7: Strict fixed-risk budget at 0.08. Coverage is reported because abstention is not success.

Method	Coverage	Succ.	Safe	Damage	FalseIso	Utility
oracle	1.000	0.880	0.132	0.054	0.002	0.596
sensor_v5	0.549	0.422	0.022	0.008	0.292	0.231
sensor_v4	0.112	0.075	0.000	0.000	0.112	0.020
risk_reconfig	0.087	0.054	0.000	0.000	0.084	0.010
moe_router	0.035	0.022	0.000	0.000	0.035	0.003
causal_graph	0.037	0.021	0.000	0.000	0.037	0.002
bayes_fusion	0.022	0.014	0.000	0.000	0.022	0.002
conformal	0.024	0.013	0.000	0.000	0.024	0.001
ind_faults	0.006	0.003	0.000	0.000	0.006	0.000
dropout_tx	0.026	0.014	0.000	0.000	0.026	-0.000
ensemble_gate	0.012	0.006	0.000	0.000	0.012	-0.001
worst_sensor	0.011	0.005	0.000	0.000	0.011	-0.002

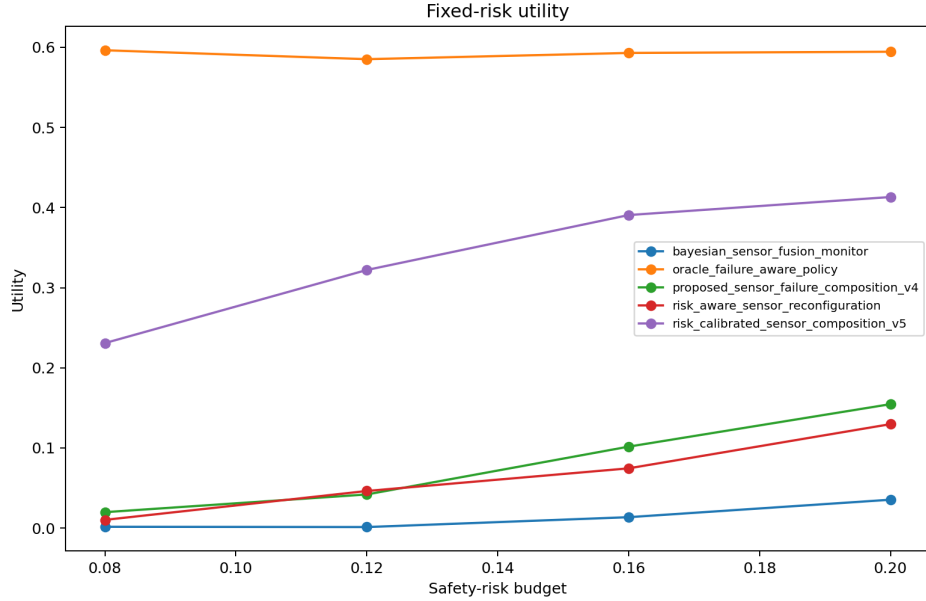


Figure 6: Fixed-risk utility over budgets. V5 keeps meaningful strict-budget coverage while simpler methods often abstain or repair too conservatively.

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